

The Potential Role of Pyrroloquinoline Quinone to Regulate Thyroid Function and Gut Microbiota Composition of Graves' Disease in Mice

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Abstract

Graves' disease (GD) is an autoimmune disorder disease, and its prevalence continues to increase worldwide. Pyrroloquinoline quinone (PQQ) is a naturally antioxidant compound in milk, vegetables, and meat. We aim to identify the treatment efficacy of PQQ on GD and its regulatory effect on intestinal microbiota. The GD mice model was built by an adenovirus expressing autoantigen thyroid-stimulating hormone receptor (Ad-TSHR289). Fecal samples were collected for 16S rDNA sequencing after PQQ pretreatments (20, 40, or 60 mg/kg BW/day) for 4 weeks. Thyroid and intestine functions were measured. The levels of serum TSHR and T4 were significantly raised, and the thyroid gland size was typically enlarged in the GD group than in controls, reversed by PQQ therapy. After PQQ replenishment, *IL6* and *TNFA* levels in small intestine tissues were lower than those in the GD group, with *Nrf2* and *HO1* levels improved. Also, the PQQ supplement could maintain the mucosal epithelial barrier impaired by GD. In microbial analyses, PQQ treatment could prompt the diversity recovery of gut microbiota and reconstruct the microbiota composition injured by GD. *Lactobacillus* served as the most abundant genus in all groups, and the abundance of *Lactobacillus* was increased in the GD group than in control and PQQ groups. Besides, *Lactobacillus* was highly correlative with all samples and the top 50 genera. PQQ supplementation regulates thyroid function and relieves intestine injury. PQQ changes the primary composition and abundance of GD's intestine microbiota by moderating *Lactobacillus*, which may exert in the pathogenesis and progression of GD.

Key words: Graves' disease, pyrroloquinoline quinone, thyroid function, gut microbiota, *Lactobacillus*

Introduction

Graves' disease (GD) is an autoimmune disorder characterized by an enlarged thyroid gland, Graves' orbitopathy (GO), an accelerated heart rate, and hyperthyroidism (Wémeau et al. 2018; Davies et al. 2020; Hoang et al. 2022). Moreover, the upregulation of thyroid-stimulating hormone receptor antibodies (TRAB) and free thyroxine (fT4) is the main immunological feature in GD (Ehlers et al. 2019; Davies

et al. 2020). In GD, the loss of immune tolerance to thyroid antigens is attributed to a complex interplay between genetic and environmental factors (Neag and Smith 2022). It is well known that T-cell-mediated immunity, TSHR, and mesenchymal stem cell properties in orbital fibroblasts have participated in the pathogenesis of GO (Taylor et al. 2020). However, the precise mechanism underlying Graves' disease (GD) remains unclear. Recently, some studies have provided new insight into gut microbiota referring to the

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pathogenesis of GD (Knezevic et al. 2020; Moshkelgosha et al. 2021).

The human body harbors trillions of microorganisms on both mucosal and epidermal surfaces, with a significant concentration in the gastrointestinal tract. The intestinal microbiota is crucial for human health and has physiological functions such as regulating nutrient metabolism, immune regulation, and protecting the intestinal structure and function (Jandhyala et al. 2015). Once the intestinal microbiota is disrupted, it can cause various diseases, such as asthma, allergies, inflammatory bowel disease (Arrieta et al. 2015; Bunyavanich et al. 2016; Nishino et al. 2018). Currently, many studies have shown an association between gut microbiota and thyroid diseases. The gut microbiota has been reported to regulate thyroid hormone metabolism by controlling the absorption and degradation of thyroid hormone iodine and the enterohepatic circulation of thyroid hormone (Lin and Zhang 2017; Fröhlich and Wahl 2019). There are also studies indicating that the imbalance of Th17 and T_{reg} cells caused by gut microbiota is related to the pathogenesis of GD (Smith et al. 2013). Thus, it can be seen that the gut microbiota is closely related to the occurrence and development of GD.

In 1964, pyrroloquinoline-quinone (PQQ) was initially identified as a bacterial dehydrogenase coenzyme and has since been associated with various biological processes such as mitochondriogenesis, growth reproduction, and aging (Hauge 1964; Mohamad Ishak and Ikemoto 2023). Due to its quinone structure, PQQ can undergo reduction by numerous substances and can be restored to its oxidized state by oxygen (Mohamad Ishak and Ikemoto 2023). A study suggests that in an osteoporosis model induced by estrogen deficiency, dietary supplementation with PQQ can inhibit oxidative stress and osteocyte senescence to alleviate bone loss effectively (Geng et al. 2019). Consequently, the supplementation of PQQ was acknowledged for its significant role in promoting longevity, as well as its antioxidant and anti-inflammatory effects (Ouchi et al. 2013; Sasakura et al. 2017; Yin et al. 2019). However, limited attention was given to the effects of PQQ treatment on GD mice and gut microbiota. Therefore, the objective of this study is to explore the potential therapy effects of PQQ in regulating GD and reconstructing intestine homeostasis.

Experimental

Materials and Methods

Animal model and dietary intervention. Eight-week-old female BALB/c mice were obtained from Beijing Vital River Laboratory Animal Technology Co., Ltd. (China). They were maintained in a specific

pathogen-free facility (Shanghai Tenth People's Hospital Animal Center, China) under standard conditions with $22 \pm 2^\circ\text{C}$, $60 \pm 5\%$ humidity and 12-hour light/dark cycles. Mice were fed with a standard diet and water. All animal experiments were approved by the Animal Care Institution and Ethics Committee of Shanghai Tenth People's Hospital (20KT127).

The construction of recombinant adenovirus Ad-TSHR289 encoding TSHR: the subunit and identification of virus transfection ability were carried out by Shanghai Jikai Gene Technology Co., Ltd. (China). All the Ad-TSHR289 were mixed with phosphate-buffered saline (PBS, pH = 7.3) in a total volume of 100 μl for immunization. Then, the prepared Ad-TSHR289 solution was injected into mice for three times at 3-week intervals.

PQQ (purity > 98%) was provided by Shanghai Maclin Biochemical Technology Co., Ltd. (China). The mice were randomly divided into six groups (5 mice per group): 1) control group (treated with PBS only); 2) GD group (treated with Ad-TSHR289 only); 3) GD + PQQ low group (treated a low concentration of PQQ, i.e., 20 mg PQQ/kg BW/day); 4) GD + PQQ mid group (treated with a middle concentration of PQQ, i.e., 40 mg PQQ/kg BW/day); 5) GD + PQQ high group (treated with a high concentration of PQQ, i.e., 60 mg PQQ/kg BW/day) PQQ was dissolved in PBS in 0.2 ml per mouse and administered orally from 4 weeks to 8 weeks after the last injection (Qu et al. 2022); 6) GD + MMI (methimazole) (Meilun Biotech Co., Ltd., China) group (treated with MMI, i.e., 2.5 mg MMI/kg BW/day). MMI was dissolved in PBS in 0.2 ml per mouse and administered orally from 4 weeks to 8 weeks after the last injection. After the final oral gavage, mice were sacrificed for further tests.

Histopathology. After the mice were euthanized, the separated thyroid and small intestine tissue were fixed with 10% buffered formalin, paraffin-embedded sections were stained with the hematoxylin and eosin (H&E) (Sigma-Aldrich, USA), and the histopathological changes, including the severity of inflammatory cell infiltration and the extent of crypt damage were observed under light microscopy to access the intestinal barrier integrity. In this study, the board-certified pathologist blinded to the experimental condition samples scored the samples.

ELISA assay. Total thyroxine (T4) and TSH receptor autoantibodies (TRAb) in mouse serum were measured using Mouse T4 Elisa Kit (YCQZ-10219; shybio, China), Mouse TRAb Elisa Kit (YCMZ-35270; shybio, China) according to the manufacturers' instructions.

Reverse transcription (RT)-quantitative PCR (RT-qPCR). Total RNA was extracted from the small intestine by using the Total RNA Isolation Kit V2 RC112 (Vazyme Biotech Co., Ltd., China). Reverse transcription

was carried out using HiScript® III All-in-one RT SuperMix Perfect for qPCR R333 (Vazyme Biotech Co., Ltd., China). RT-qPCR was performed on the LightCycler® 96 System (Roche Diagnostics (Shanghai) Ltd., China) using Taq Pro Universal SYBR qPCR Master Mix Q712 (Vazyme Biotech Co., Ltd., China). The relative expression of mRNAs was performed by the $2^{-\Delta\Delta Ct}$ method using GAPDH mRNA for normalization. The primer information is shown in Table SI.

Fecal sample collection, DNA extraction, and sequencing. Fecal samples of mice were collected after the final oral gavage, snap-frozen in liquid nitrogen, and stored at -80°C until use. The DNA of total bacteria in mice feces was extracted with the E.Z.N.A.® Soil DNA Kit (Omega Bio-Tek, Inc., USA) according to the manufacturer's protocols. The purity and concentration of DNA were detected by NanoDrop™ 2000 (Thermo Fisher Scientific Inc., USA). The thermocycler polymerase chain reaction system GeneAmp™ 9700 (Applied Biosystems, USA) was utilized to amplify the bacterial 16S ribosomal RNA (rRNA) genes hypervariable region 3 (V3) and 4 (V4) with the primers 338F (5'-ACTCCTACGGGAGGCAGCAG-3') and 806R (5'-GGACTACHVGGGTWTCTAAT-3'). To ensure the accuracy of the subsequent data analysis, each sample was amplified three times. The products were further purified using the Axygen® AxyPrep DNA Gel Extraction Kit (Corning, USA), and quantitative analysis was carried out using QuantiFluor™-ST Fluorometer (Promega Corporation, USA), according to the manufacturer's protocol. Sequencing was performed on an Illumina® MiSeq™ platform (Illumina, Inc., USA) by combining the purified amplicons at equimolar concentrations and performing end pair sequencing analysis.

Analysis of 16S rRNA gene sequencing. The analysis of 16S rRNA gene sequencing data was performed in the Majorbio Cloud Platform (<https://cloud.majorbio.com>). Paired-end reads were first assembled using FLASH (v1.2.11) (Magoč and Salzberg 2011). The sequencing data were then trimmed, filtered, and clustered into operational taxonomic units (OTUs) using a 97% similarity cutoff with UPARSE (v7.1) (Edgar 2013). All sequence classification methods were annotated using an RDP classifier (v2.2) (Wang et al. 2007) combined with the Silva database, with a confidence threshold of 70%. OTUs with $<0.005\%$ of sequences were removed to decrease the effect of spurious sequences on the results.

Statistical analyses. This study presents the data as means $\pm$ SEM. To assess significance, the mean values of independent groups were compared using Student's *t*-test, One way ANOVA, and Wilcoxon rank sum test as follows: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. Results with $p < 0.05$ were considered statistically significant.

Results

PQQ supplement modulates thyroid function and ameliorates thyroid injury. The GD model in mice was initially established using Ad-TSHR289, and its validity was confirmed through the measurement of serum TSHR and T4 levels using Elisa, as well as the examination of the thyroid gland using HE staining (Fig. 1). Our findings revealed a significant increase in serum TSHR and T4 levels (Fig. 1A and 1B), along with a noticeable enlargement of the thyroid gland (Fig. 1C) in the GD group compared to the controls. Subsequently, we evaluated the therapeutic effects of PQQ at varying concentrations, namely low, middle, and high. The results showed that serum TSHR and T4 were remarkably decreased (Fig. 1A and 1B). The size of the thyroid gland was typically reduced (Fig. 1C) in three cohorts of PQQ replenishment compared with the GD group, suggesting the significant effect of PQQ treatment on GD, among which middle concentration group demonstrated the best efficacy (Fig. 1).

PQQ supplement decreases inflammation and oxidative stress response and alleviates the small intestine epithelial injury. Alterations in the thyroid-gut axis are considered one of the modern views of GD pathogenesis (Knezevic et al. 2020). Consequently, our study aimed to investigate the impact of PQQ treatment on the small intestine system. In the GD model, *IL6* and *TNFA* levels in small intestine tissues were firmly higher than in controls, with *Nrf2* and *HO1* levels lower, implying high inflammation and oxidative stress levels in the GD enteric tract (Fig. 2A and 2B). Following PQQ therapy at three different concentrations, the levels of *IL6* and *TNFA* in small intestine tissues were significantly reduced compared to the GD group. Additionally, the levels of *Nrf2* and *HO1* were elevated, suggesting that PQQ effectively mitigated inflammation and oxidative stress in the small intestine (Fig. 2A and 2B). Furthermore, our findings demonstrated that the intestinal epithelial barrier was noticeably compromised in the GD group. In contrast, supplementation with PQQ facilitated the preservation of the mucosal epithelial barrier, as confirmed by H&E staining (Fig. 2C).

Oral gavage of PQQ restores the diversity of gut microbiota in the GD group. In our previous studies, we have investigated the close relationship between GD and gut microbiota (Jiang et al. 2021). Given the significant efficacy of PQQ at moderate concentrations, we were particularly interested in examining the impact of PQQ with a moderate concentration on the alteration of gut microbiota in GD. The analysis of rank-abundance curves and rarefaction curves of the Sobs index at the OTU level revealed that the species in the three groups exhibited a uniform and abundant distribution, with the sequencing data being sufficient

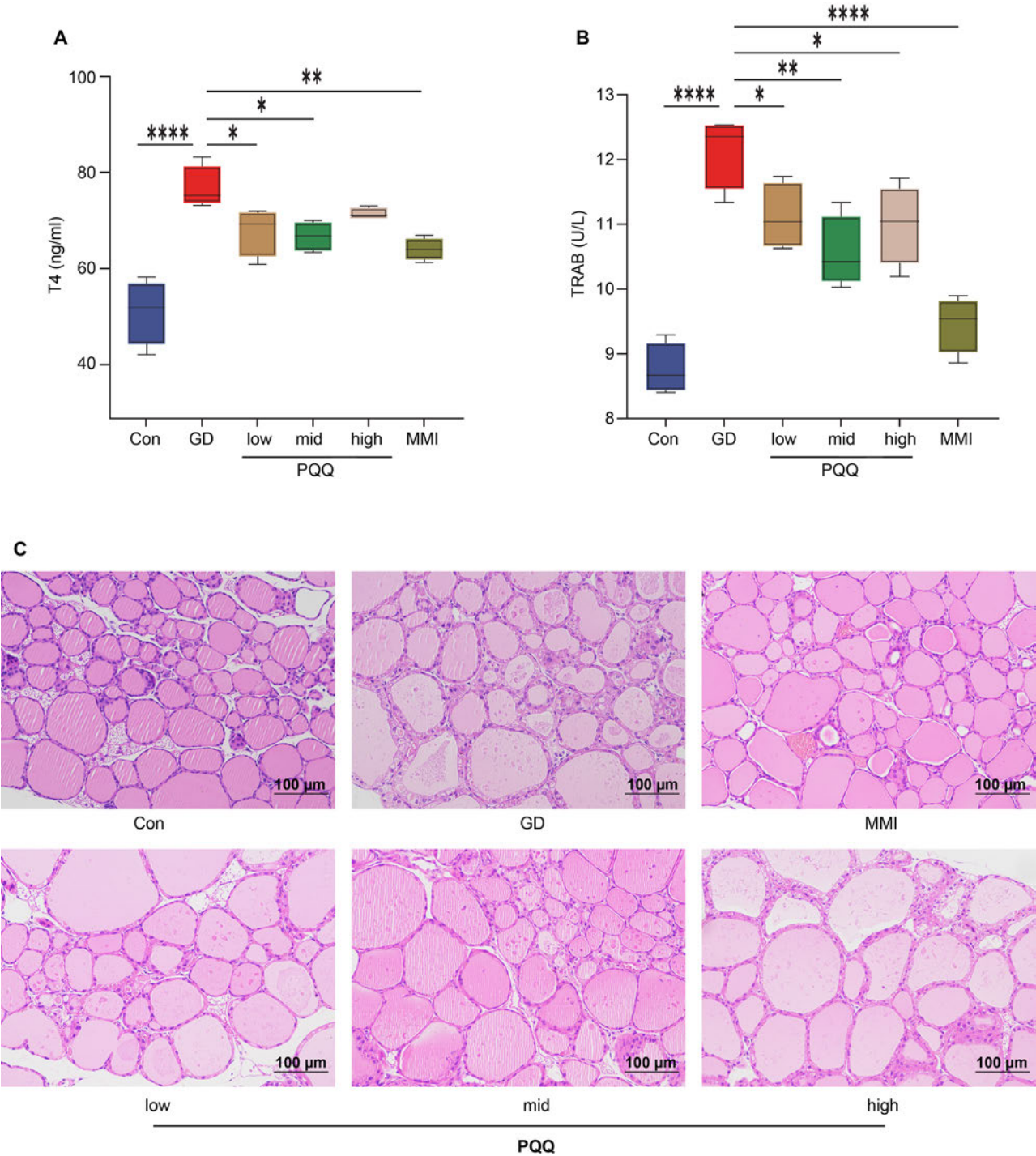
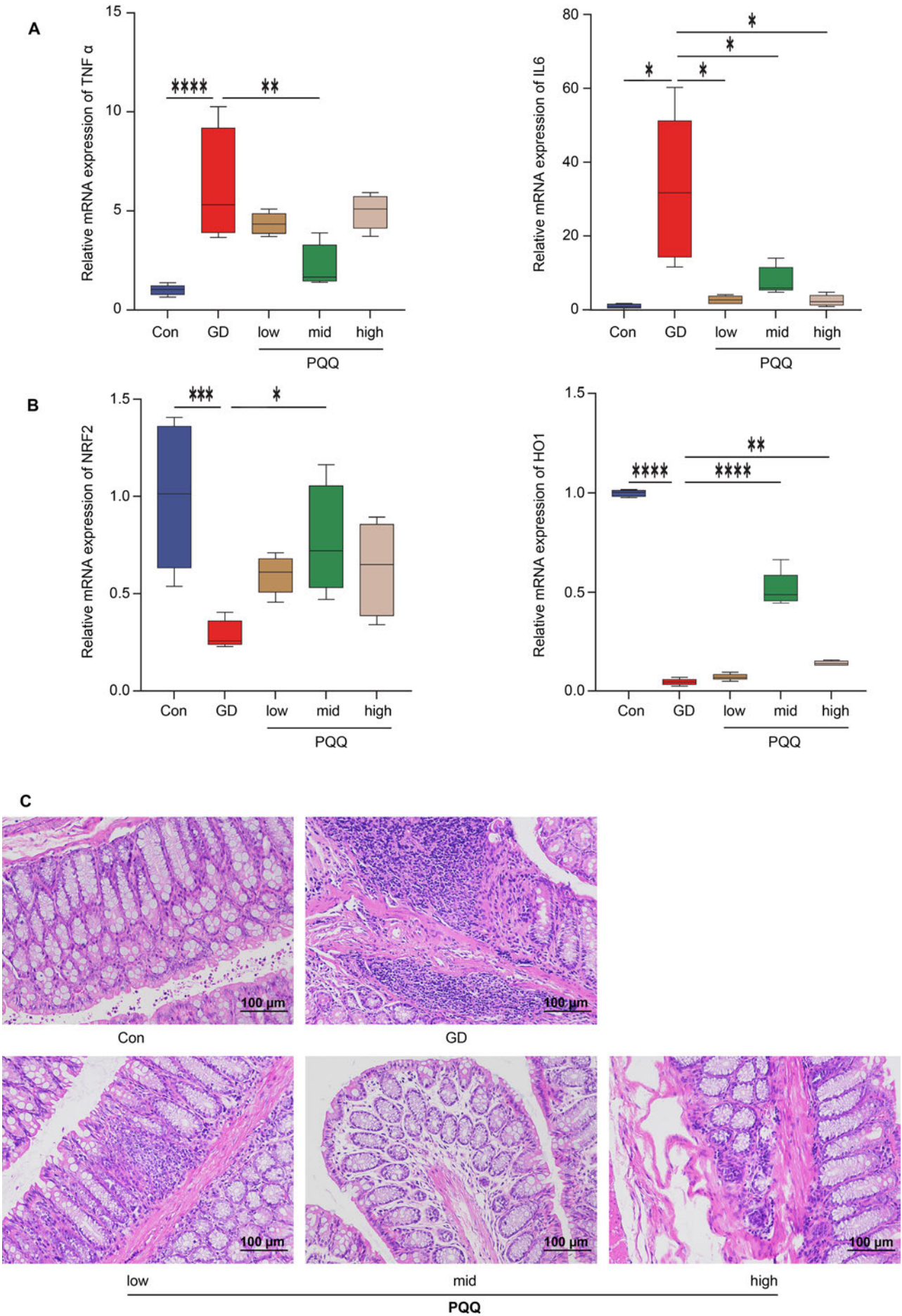


Fig. 1. PQQ supplement modulates thyroid function and ameliorates thyroid injury. A) The concentrations of T4 and B) TRAb levels were determined inserum samples using Mouse ELISA Kit, C) the representative pictures of H&E staining of thyroids (200× magnification with 100 μm of scale). Data were represented as mean ± SEM, **p* < 0.05, ***p* < 0.01, ****p* < 0.001 between groups. Con – control group, GD – Graves' disease group, GD + PQQ/low – GD mice treated with 20 mg PQQ/kg BW/day, GD + PQQ/mid – GD mice treated with 40 mg PQQ/kg BW/day, GD + PQQ/high – GD mice treated with 60 mg PQQ/kg BW/day, GD + MMI – GD mice treated with 2.5 mg MMI/kg/BW/day, PQQ – pyrroloquinoline quinone, MMI – methimazole

Fig. 2. PQQ supplement decreases inflammation and oxidative stress response and alleviates small intestine epithelial injury. Relative expression of mRNA in colons of A) *IL6*, *TNFα*, B) *Nrf2*, *HO1* was detected by PCR, C) the representative pictures of the hematoxylin and eosin (H&E) staining of colons (200× magnification with 100 μm of scale). Data are represented as mean ± SEM, **p* < 0.05, ***p* < 0.01, ****p* < 0.001 between groups



in quantity (Fig. 3A and 3B). Specifically, there were a total of 15 phyla, 25 classes, 57 orders, 86 families, 162 genera, and 236 species, and 739 OTU in the controls; 9 phyla, 15 classes, 36 orders, 55 families, 116 genera, 183 species and 667 OTU in the GD group; and 10 phyla, 14 classes, 44 orders, 78 families, 163 genera, 266 species, and 1045 OTU were in the PQQ group. By examining α diversity using Sobs, Ace, and Chao indices at the OTU level, it was observed that the diversity and abundance were significantly reduced in the GD group than controls and greatly improved in the PQQ

group (Fig. 3C, 3D and 3E). These findings suggest that the administration of PQQ could effectively restore the diversity of intestinal flora that GD has disrupted. In the analysis of β diversity, including principal component analysis (PCA), principal coordinate analysis (PCoA) based on bray_curtis, binary_jaccard, and unweighted_unifrac distance, as well as non-metric multidimensional scaling (NMDS) at the genus level, it was evident that the bacterial composition in the GD group differed significantly from that of the control and PQQ groups (Fig. 3F–3J). In contrast, the microbial structure

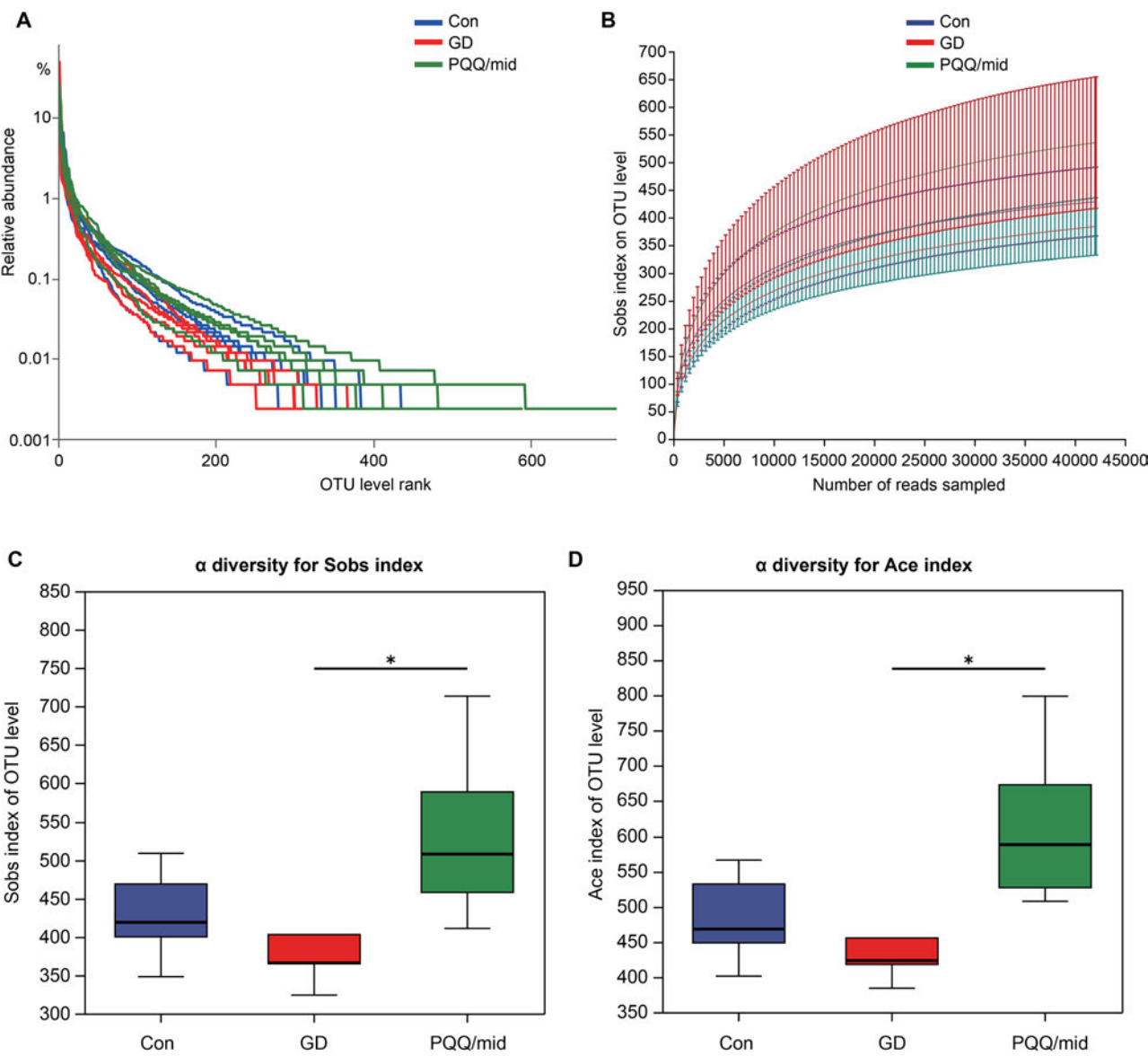
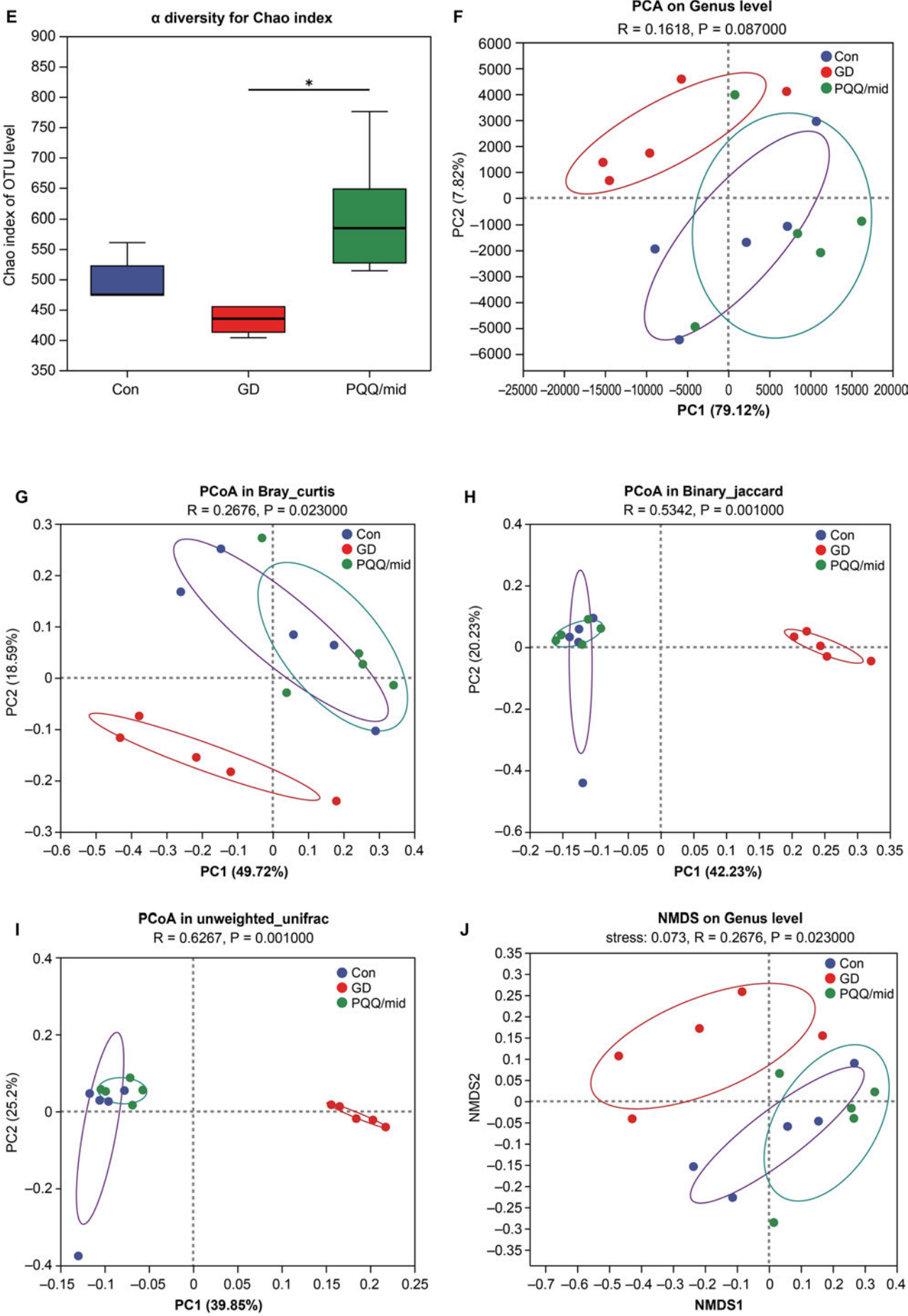


Fig. 3. Oral gavage of PQQ restores the diversity of gut microbiota in the GD group.

A) Rank-abundance curves of the gut microbiota at the OTU level, B) rarefaction curves (SOBs index) of the gut microbiota at the OTU level. Comparison of α diversity in C) Sobs, D) Ace, E) Chao indices in Con, GD and PQQ/mid groups at the OTU level. Analysis of β diversity using F) PCA, G) PCoA based on bray_curtis, H) binary_jaccard and I) unweighted_unifrac distance and J) NMDS on the genus level.

** $p < 0.01$; *** $p < 0.001$. OTUs – operational taxonomic units, PCA – principal component analysis, PCoA – principal coordinate analysis, NMDS – non-metric multidimensional scaling



exhibited a high degree of similarity between the control and PQQ groups (Fig. 3F–3J). The above results demonstrated that the PQQ supplement could reconstruct the microbiota composition influenced by GD.

PQQ alters the composition and abundance of intestine flora in the GD group. The taxon compositions of the three groups were determined based on the genus level using a Venn diagram (Fig. 4A). The results indicated that *Lactobacillus* was the most abundant genus (37.12%) among the three groups (Fig. 4D), suggesting its potential significance in microbial functions. *Akkermansia*, on the other hand, was exclusively found in the control group with a relative high abundance (16.10%) (Fig. 4B), indicating its possible absence in the progression of GD. Additionally, *unclassified_f_Enterobacteriaceae* (57.14%) and *Odoribacter* (46.74%) were identified as the most abundant genera in the GD and PQQ groups, respectively (Fig. 4C and 4E). Community abundance on phylum level showed that Firmicutes and Bacteroidota were recognized as the two dominant phyla among all groups (Fig. 4F). The abundance of Firmicutes was decreased in the GD group than in controls, further reduced in the PQQ group; in contrast, the abundance of Bacteroidota was increased in the GD group than controls, further improved in the PQQ group (Fig. 4F). On the genus level, *Lactobacillus* and *Lachnospiraceae_NK4A136_group* owned the most abundance among three groups (Fig. 4G–4J). Interestingly, the abundance of *Lactobacillus* was increased in the GD group than in control and the PQQ groups; conversely, the abundance of *Lachnospiraceae_NK4A136_group* was decreased in the GD group than in control and PQQ groups. The heatmap on the abundance of the top 30 genera was presented in Fig. 4K, where the *Lactobacillus* owned the predominate propriety. Furthermore, the Circos analysis (Krzywinski et al. 2009) demonstrated the abundance association between the sample and the top 10 generations (Fig. 4L).

Spearman correlation analysis and PICRUSt prediction under PQQ supplement in the GD group. Differential distributed genera validated by the Kruskal-Wallis *H* test were shown in Fig. 5A and 5C, and *Lactobacillus* was significantly increased in the GD group than others. Furthermore, the Linear discriminant analysis Effect Size (LEfSe) (Segata et al. 2011) analysis identified 14 specific genera that displayed noticeable differences across all groups (Fig. 5B). Notably, *Lactobacillus* emerged as the most dominant genus in the GD group, and the PQQ group exhibited 11 dominant genera (Fig. 5B). Heatmap of Kyoto Encyclopedia of Genes and Genomes (KEGG) (Kanehisa and Goto 2000; Kanehisa et al. 2023) pathway based on the PICRUSt (Langille et al. 2013) prediction among three groups demonstrated that the three most prominent pathways were “Metabolic pathways”, “Biosynthesis of

secondary metabolites”, and “Microbial metabolism in diverse environments” (Fig. 5D). Additionally, the distribution of Clusters of Orthologous Groups (COG) functional abundance, also based on the PICRUSt prediction among the three groups, highlighted that “S: Function unknown”, “J: Translation, ribosomal structure and biogenesis”, and “E: Amino acid transport and metabolism” exhibited the highest levels of abundance (Fig. 5E). Subsequently, the experiment data was analyzed using the redundancy analysis (RDA) technique, revealing the indispensable role of HO1 and TNF α in the distribution of the samples (Fig. 5F). Furthermore, Collinearity Analysis and Spearman Correlation Analysis were conducted to elucidate the interrelationships among samples within the three groups and genera, as well as the associations within the top 50 genera (Fig. 5G and 5H). The outcomes indicated a strong correlation between *Lactobacillus* and all three groups, as well as with other genera.

Discussion

The etiology of GD might be intricately linked to autoimmunity and oxidative stress (Hoang et al. 2022). However, the specific role of PQQ, a natural antioxidant compound, in GD mice remains unclear. In our study, we sought to ascertain the impact of PQQ supplementation on GD mice and their gut microbiota, and the key findings were as follows: i) The serum TSHR and T4 were decreased and the thyroid gland size was typically reduced under PQQ supplement in GD. ii) PQQ reduced inflammation and oxidative stress response and attenuated small intestine epithelial injury. iii) PQQ played a significant role in restoring the diversity and composition of microbiota. iv) Differential distributed genera validated by the Kruskal-Wallis *H* test and the LEfSe analysis showed that PQQ treatment could decrease the abundance of *Lactobacillus* compared with the GD group in mice, a potential target therapy for GD processes.

In the initial phase of our study, we assessed the influence of different concentrations of PQQ on GD. It was observed that low, middle, and high concentrations of PQQ all demonstrated a beneficial therapeutic impact on thyroid function, with the middle concentrations yielding the most favorable outcomes. Furthermore, Kumar et al. (2014) documented the alleviation of detrimental effects, such as increased serum T3 and T4 levels, in hyperthyroidism rats after administering PQQ. Our findings further verified the PQQ’s repair results on the thyroid gland. Following this, we assessed the potential impacts of PQQ on the intestinal system. Surprisingly, PQQ demonstrated the capacity to not only regulate thyroid autoimmune conditions but also

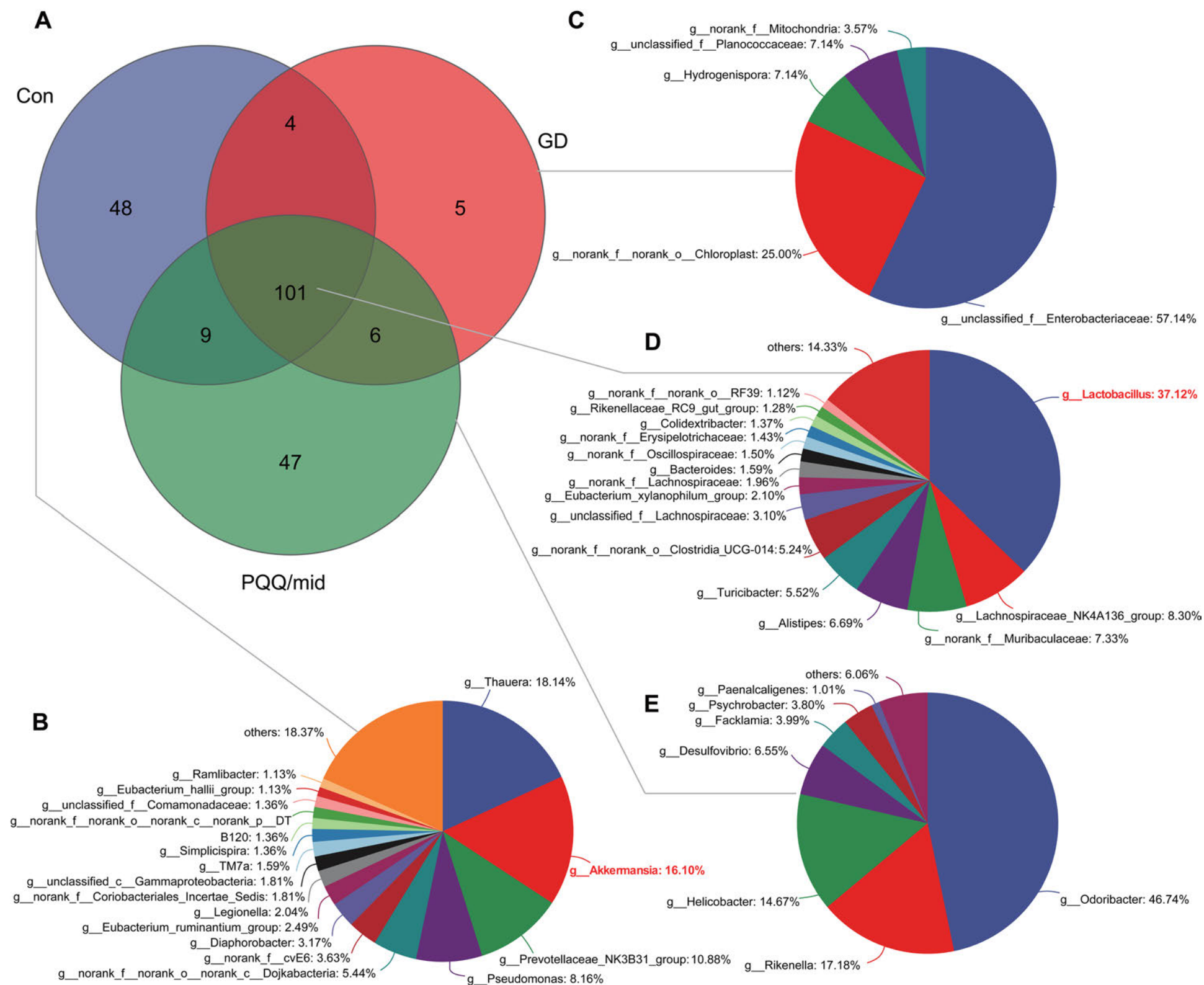


Fig. 4. PQQ alters the composition and abundance of intestine flora in the GD group.

A) The species Venn diagram at the genus level. Overlapping parts in the figure indicate common species. On the genus level, percent of community abundance unique to B) the Con group, C) GD group and E) PQQ group. D) Percent of community abundance shared by 3 groups.

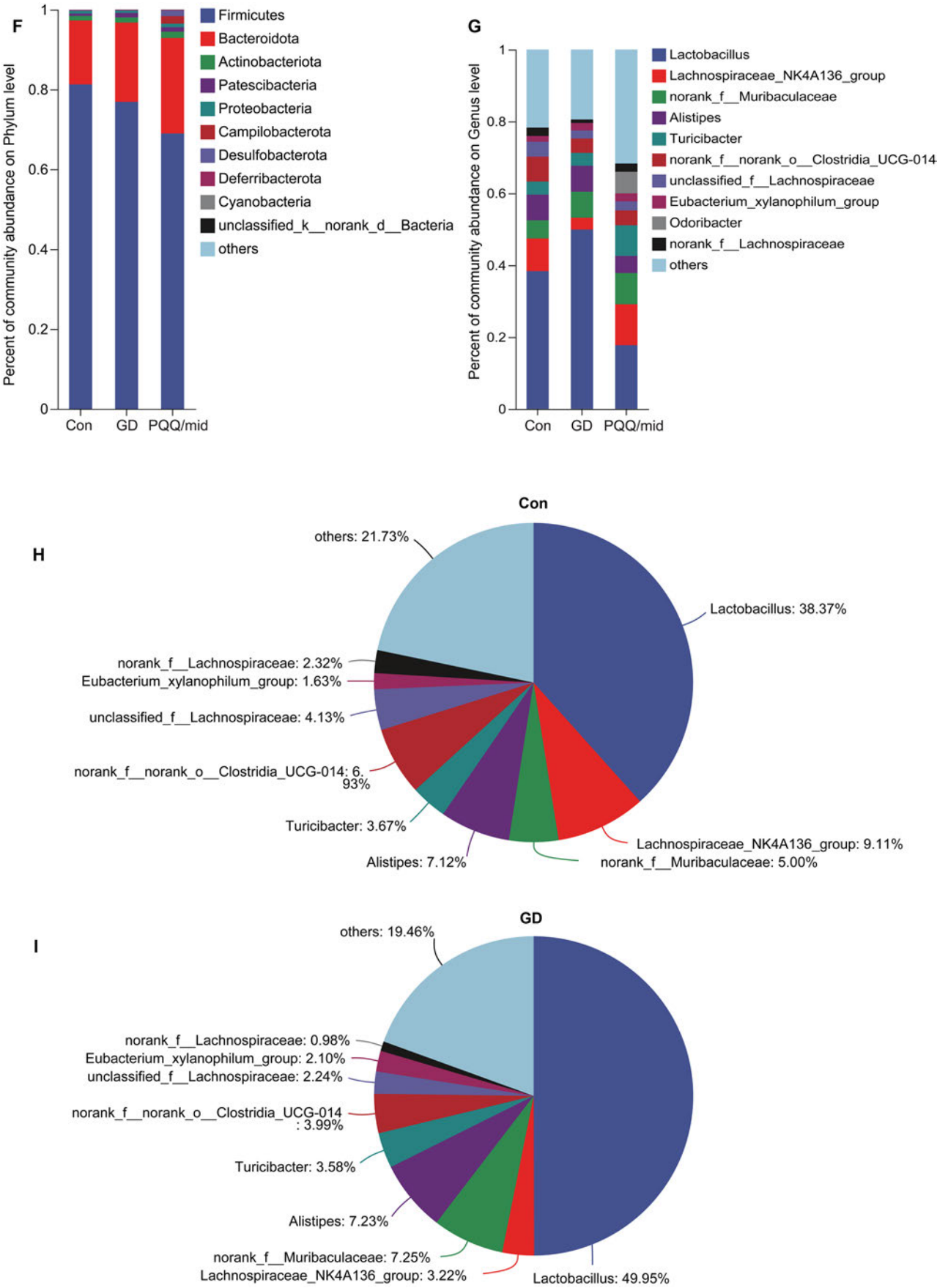
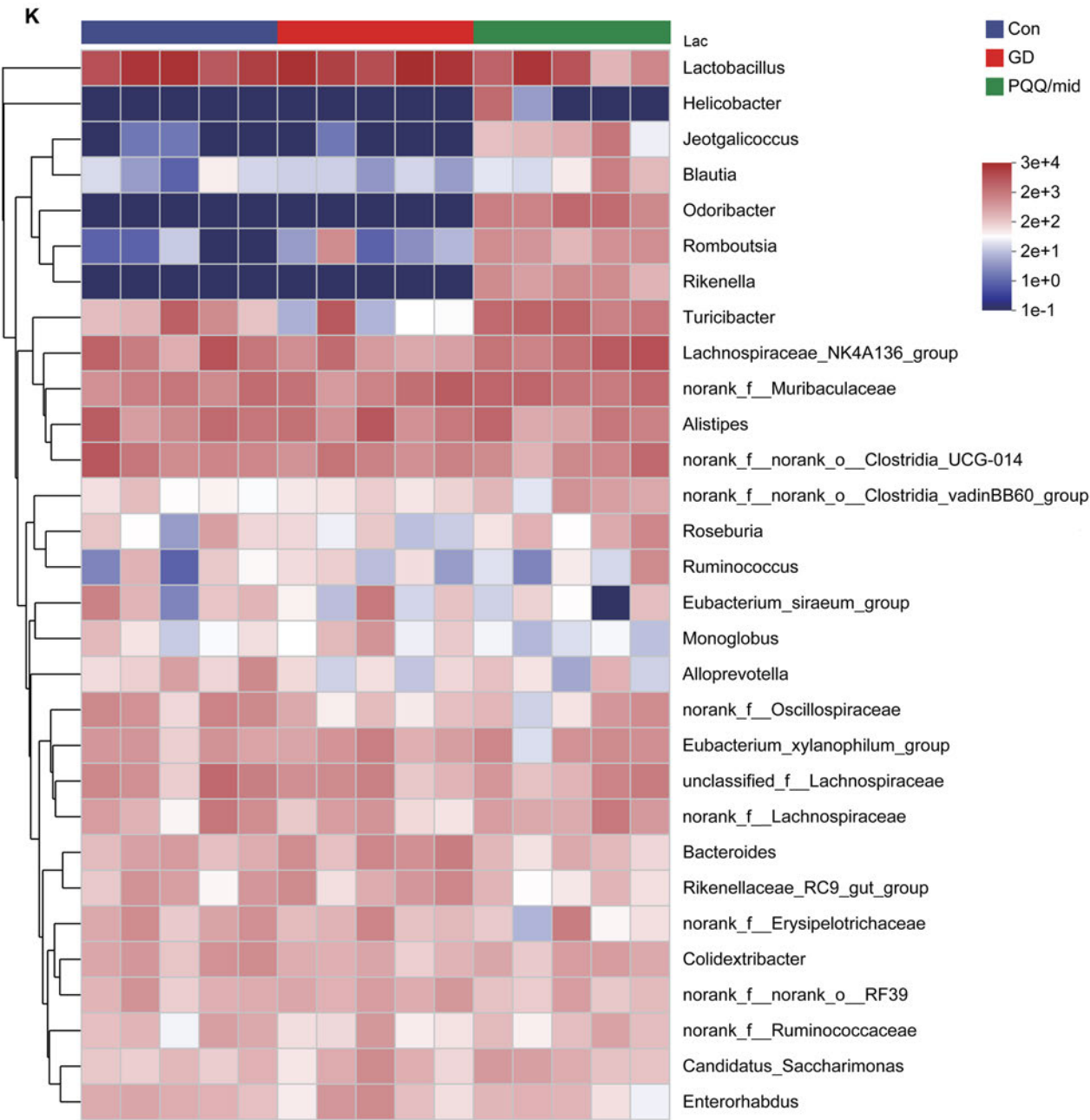
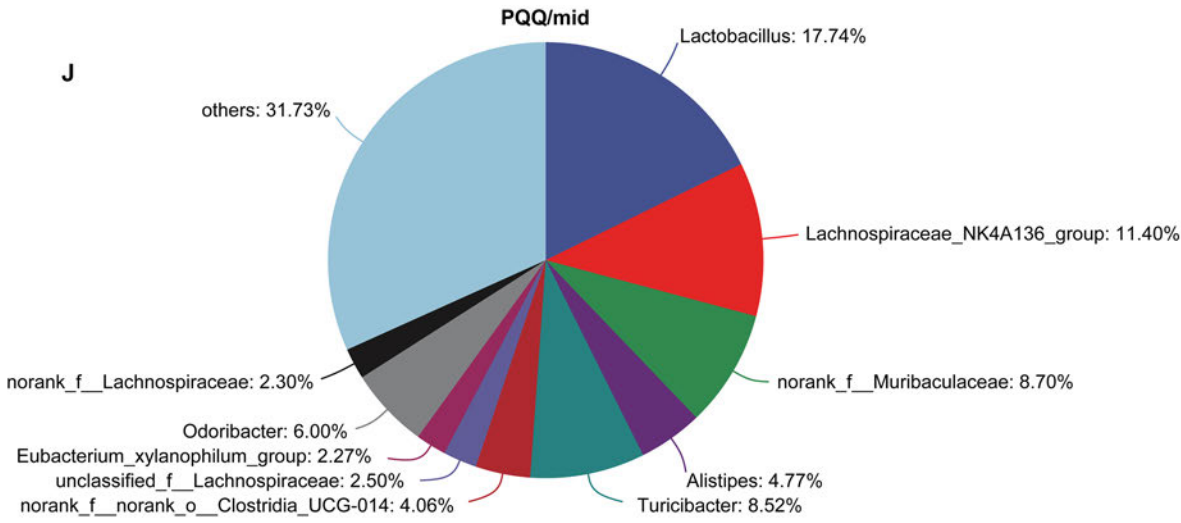


Fig. 4. PQQ alters the composition and abundance of intestine flora in the GD group.

F) Percent of community abundance of 3 groups on the Phylum level. G) Percent of community abundance of 3 groups on the Genus level. On the genus level, percent of community abundance in H) the Con group, I) GD group and J) PQQ group.



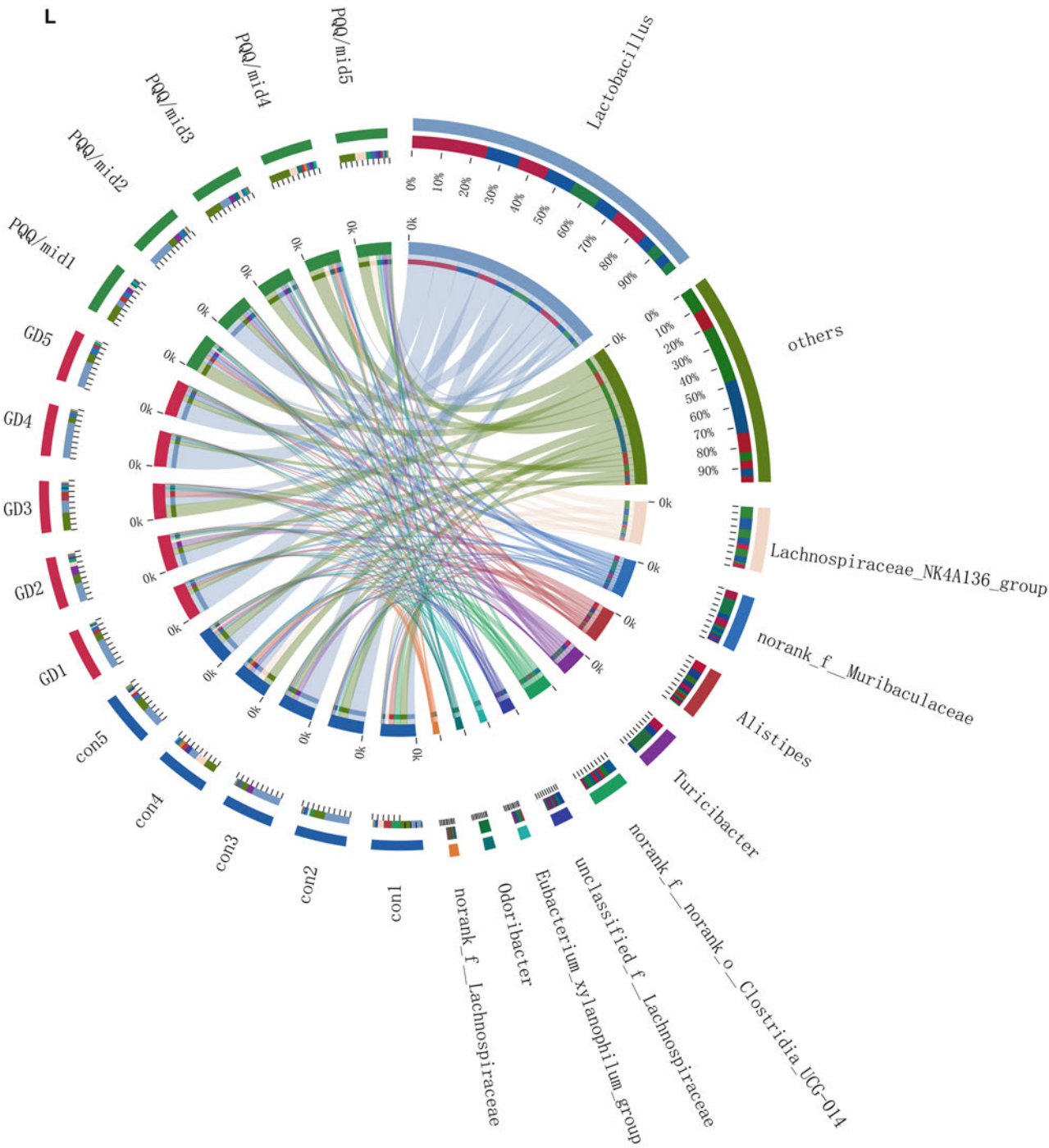


Fig. 4. PQQ alters the composition and abundance of intestine flora in the GD group.
I) GD group and J) PQQ group. K) The heatmap on the abundance of top 30 genera among 3 groups.
L) The Circos analysis of the abundance association between the sample and the top 10 genera.

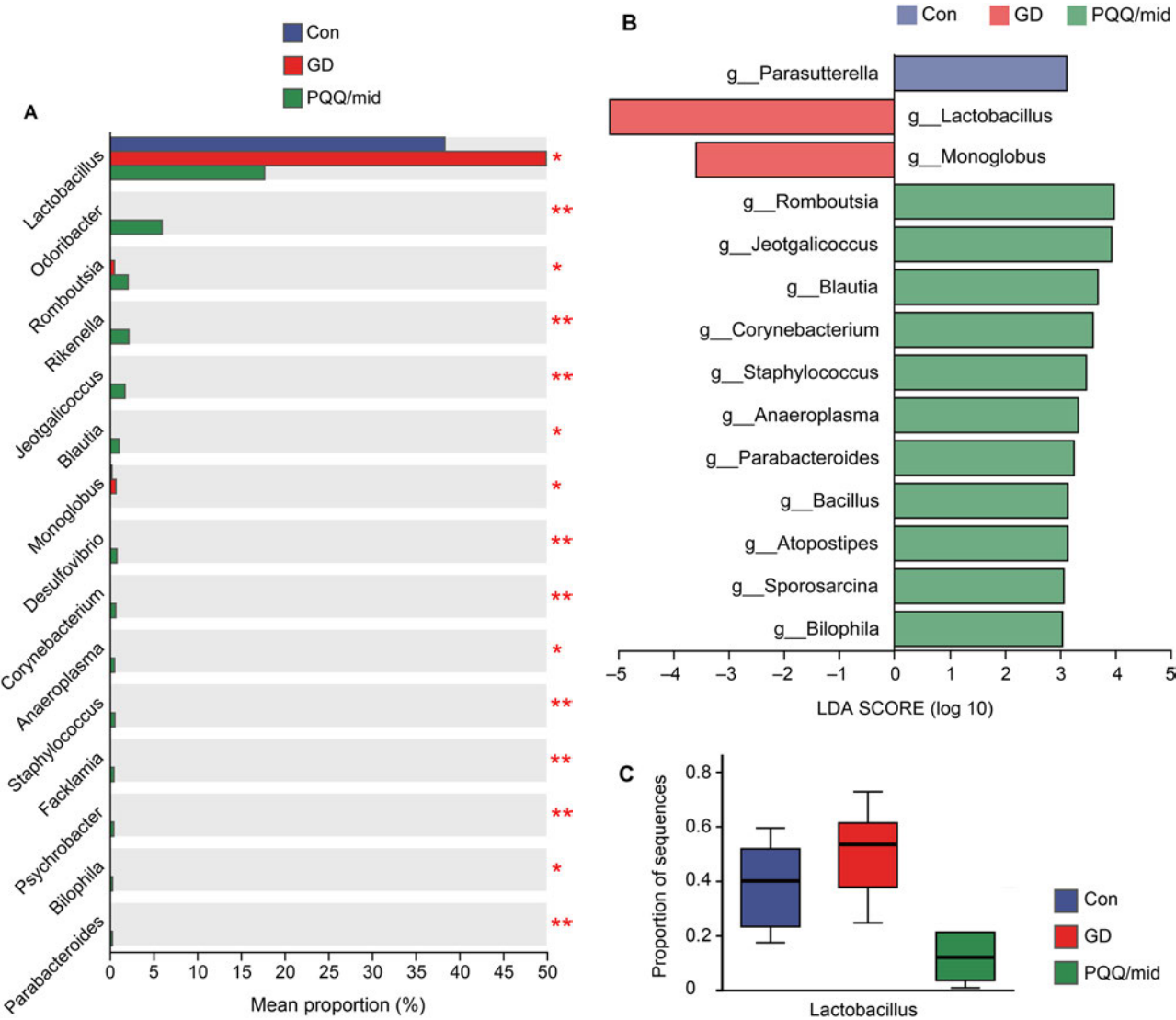


Fig. 5. Spearman correlation analysis and PICRUSt prediction under PQQ supplement in the GD group.

A) Kruskal-Wallis *H* test validated the differential distributed genera and
C) the abundance of *Lactobacillus* among all 3 groups.
B) LEfSe analysis showed that the relative abundance of 14 genera was significantly different among all groups (LDA score > 3.0 or < -3.0, *p* < 0.05).

LEfSe – linear discriminant analysis effect size, LDA – linear discriminant analysis,
KEGG – Kyoto Encyclopedia of Genes and Genomes, COG – Clusters of Orthologous Groups, RDA – redundancy analysis

alleviate inflammation and oxidative stress within the barrier of the intestinal tract. It is noteworthy that the maintenance of an intact intestinal barrier is currently acknowledged as a pivotal factor in overall health (Assimakopoulos et al. 2018). Recent research showed that impaired barrier function and microbial dysbiosis are connective with gastrointestinal disease, autoimmune disease, and a broken metabolic status in the host represented (Aldars-García et al. 2021; Fuhri Snethlage et al. 2021; Jonscher et al. 2021). In some animal models, the exposure to PQQ improved tight junction protein expression and increased jejunal barrier function, demonstrating that PQQ might act through the intestine to affect peripheral tissues (Friedman et al. 2018; Yin et al.

2019; Huang et al. 2020). Similarly, we also defined the resistance function of PQQ on the impaired gut barrier. More experiments are needed to associate thyroid autoimmune with a destructed intestinal mucosal barrier. We next performed the 16S rDNA sequencings and conducted the microbial analyses for GD under PQQ replenishment. In our previous study, we reported that a diversity on the OTU level in Ace, Sobs, Chao, Simpson, Shannon, and Coverage indices were decreased in the GD group compared to controls (Jiang et al. 2021). In this study, we further validated this conclusion and found the potential of PQQ in improving the α diversity affected by GD. In the piglet model, the Shannon index of the PQQ group was higher than that of the

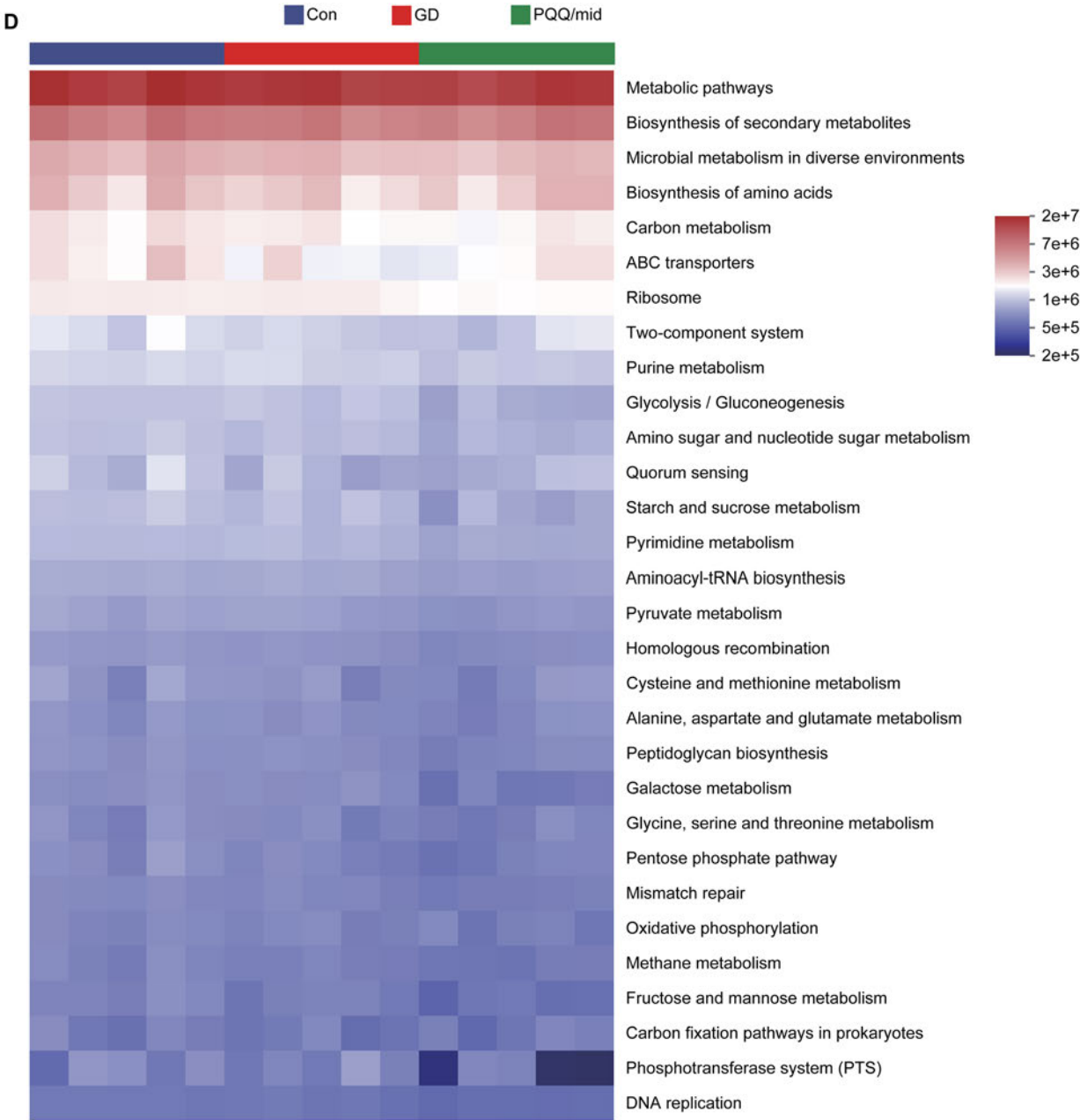
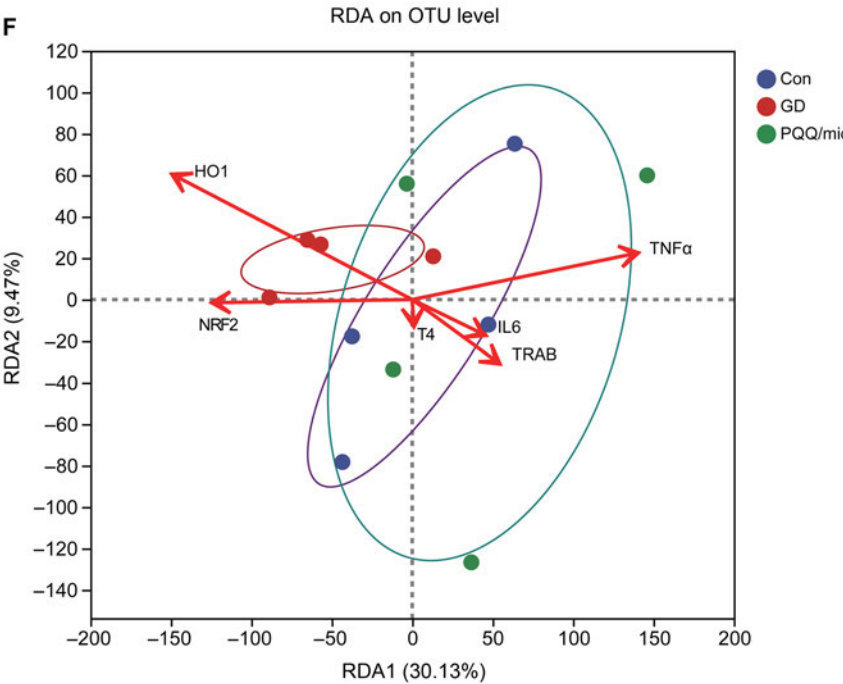
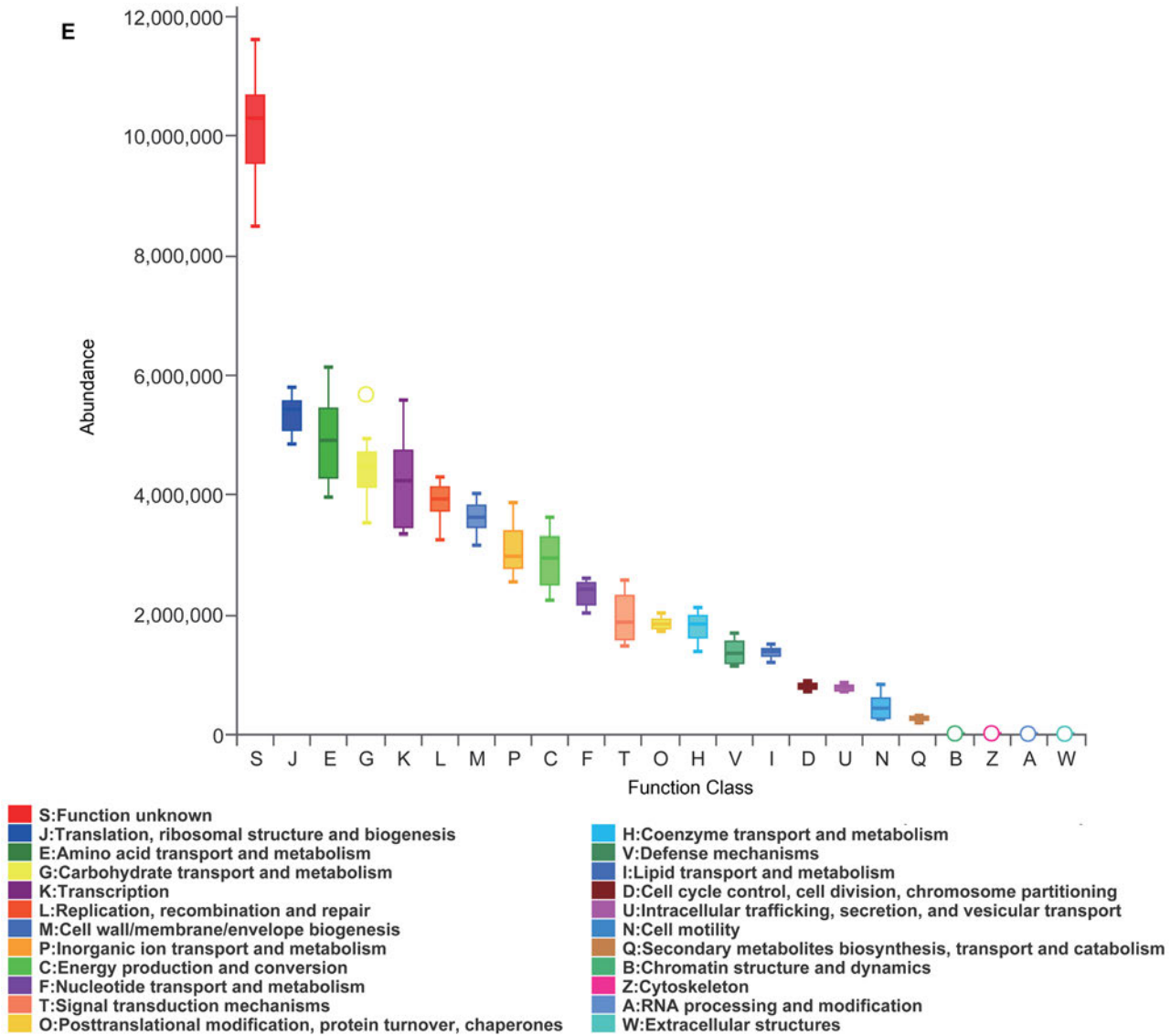


Fig. 5. Spearman correlation analysis and PICRUSt prediction under PQQ supplement in the GD group.
D) The Heatmap of KEGG pathway, and E) the COG functional abundance box distribution based on the PICRUSt prediction among 3 groups.

LEfSe – linear discriminant analysis effect size, LDA – linear discriminant analysis, KEGG – Kyoto Encyclopedia of Genes and Genomes, COG – Clusters of Orthologous Groups, RDA – redundancy analysis

controls (Huang et al. 2020). The findings of this study indicated that the administration of PQQ could potentially restore the abundance of specific microbial genera. The β diversity analysis revealed significant alterations in bacterial distribution due to the GD. However, PQQ could preserve the original composition of the gut microbiota. Similar results were observed in studies conducted on piglet and obese mice models (Friedman et al. 2018; Huang et al. 2020), suggesting that the protective effects of PQQ might be attributed to its ability to maintain the stability of the gut microbiota.

On the phylum level, the abundance of Firmicutes and *Bacteroidetes* was highest among the three groups. The proportion of Firmicutes was reduced in the GD group than controls, further decreased in the PQQ group; conversely, the proportion of Bacteroidota was elevated in the GD group than in controls, and further increased in the PQQ group, consistent with our previous study (Jiang et al. 2021). Intriguingly, prior research has shown that obese people obtain a greater abundance of Firmicutes, but lean people own a more significant proportion of abundance (Magne et al. 2020).



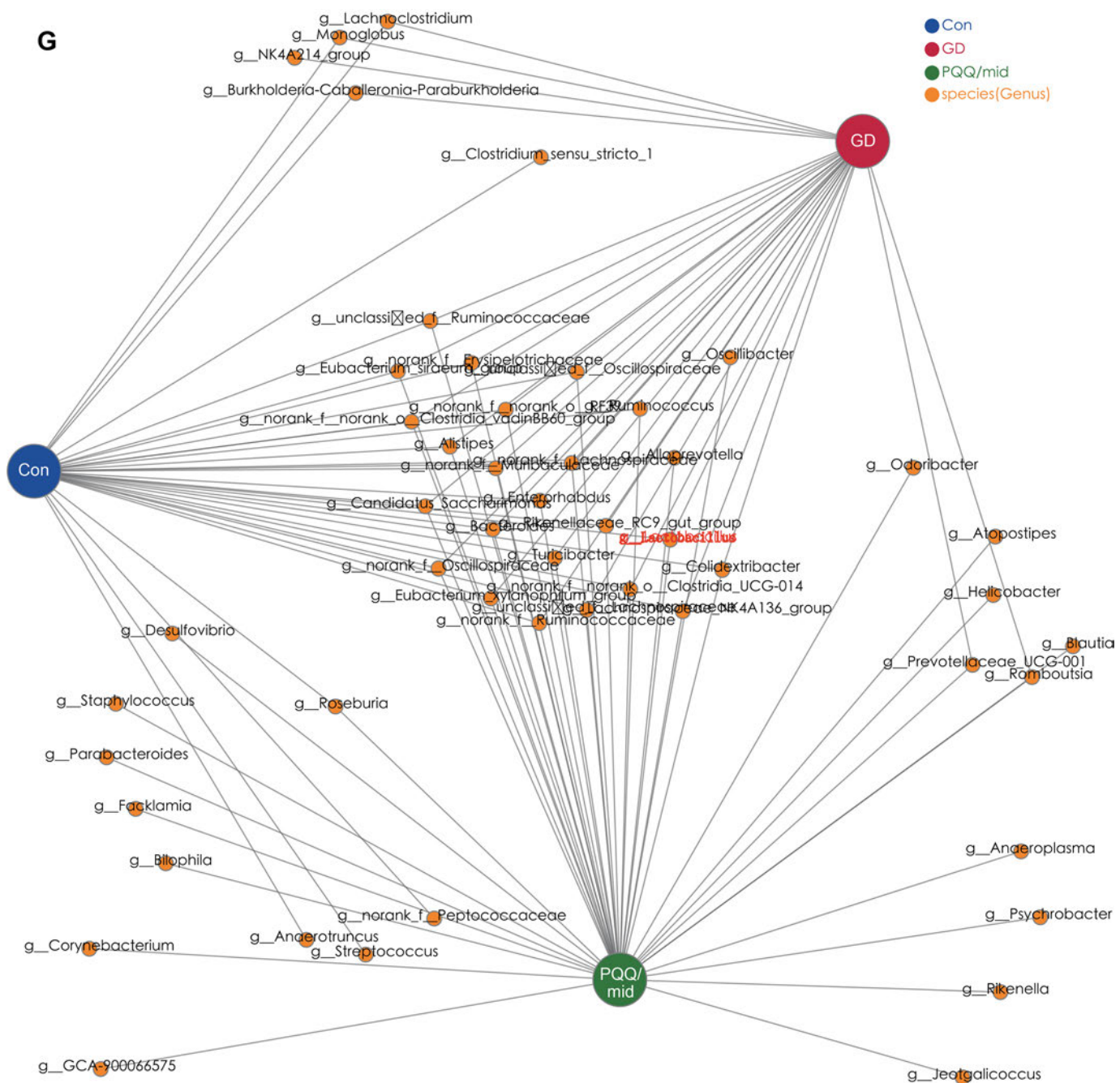


Fig. 5. Spearman correlation analysis and PICRUSt prediction under PQQ supplement in the GD group.

D) The Heatmap of KEGG pathway, and E) the COG functional abundance box distribution based on the PICRUSt prediction among 3 groups.
F) RDA results of gut microbiota and environmental factors (T4, TRAB, *IL6*, *TNFα*, *Nrf2* and *HO1*) of mice in 3 groups.
G) Network Diagram based on the Collinearity Analysis in 3 groups.

LEfSe – linear discriminant analysis effect size, LDA – linear discriminant analysis, KEGG – Kyoto Encyclopedia of Genes and Genomes, COG – Clusters of Orthologous Groups, RDA – redundancy analysis

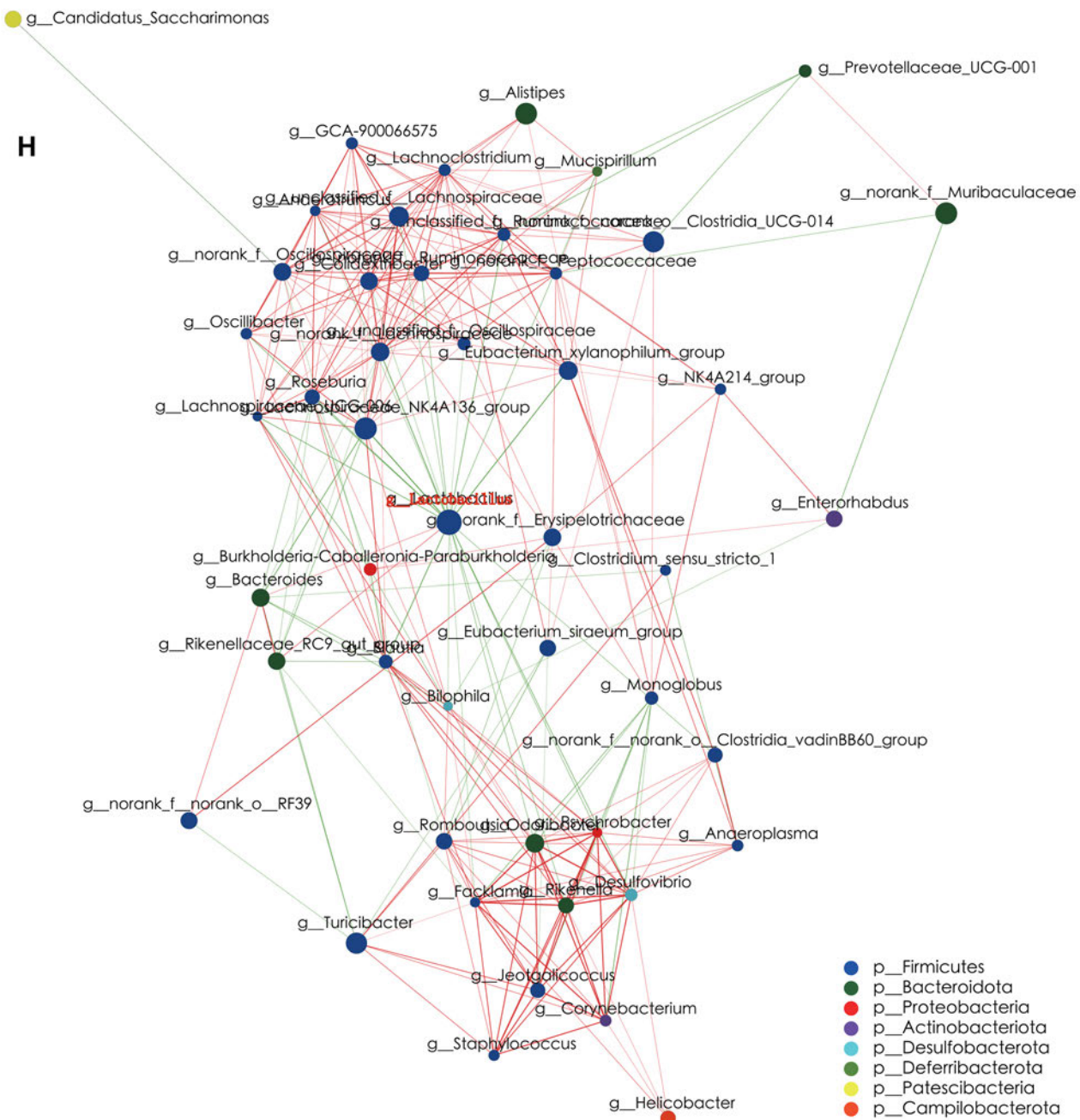


Fig. 5. Spearman correlation analysis and PICRUST prediction under PQQ supplement in the GD group.

H) Spearman Correlation Analysis between samples and top 50 genera.

LEfSe – linear discriminant analysis effect size, LDA – linear discriminant analysis, KEGG – Kyoto Encyclopedia of Genes and Genomes, COG – Clusters of Orthologous Groups, RDA – redundancy analysis

Therefore, it is hypothesized that besides the heightened basal metabolic rate, the thyroid hormones may also influence the composition and functionality of the microbiota, thereby contributing to alterations in weight. On the genus level, the abundance of *Lactobacillus* was significantly increased in the GD group than in the control and PQQ groups. *Lactobacillus* was the most primary microbiota in whole groups. This finding is further supported by the LEfSe results, which identify *Lactobacillus* as the dominant genus in the GD group. The above results indicate that *Lactobacillus*

may be involved in the pathogenesis and progression of GD, similar to the previous research in autoimmune hepatitis showing improved levels of *Lactobacillus* (Wei et al. 2020). Despite the probiotic, the study has shown that some strains of *Lactobacillus* are likely pathogenic (Sherid et al. 2016). Additionally, Miettinen et al. (2000) reported that *Lactobacillus* could directly impact the NF- κ B signaling pathway. Our results showed that PQQ treatment could decrease the abundance of *Lactobacillus* compared with the GD group in the mice, a potential target therapy for GD processes.

In our study, the KEGG database was utilized to assess the functions of distinct genera exclusive to the control, GD, and PQQ groups. We found that the “Metabolic pathway” and “Biosynthesis of secondary metabolites” exhibited in the predicted results, suggesting that these pathways were highly associated with the occurrence of controls, GD, and PQQ groups. In addition, the COG database was also involved in predicting the function in three groups. We found that the top function prediction in the three groups was characterized as “S: Function unknown”. The findings imply that there might still be an unexplored metabolic pathway in the progression of GD, and it should be the target of future research. Meanwhile, Collinearity Analysis and Spearman Correlation Analysis showed that *Lactobacillus* was highly connective with whole groups and other primary genera, and we need to pay more attention to the interaction between PQQ and *Lactobacillus* in GD.

Our research has several limitations that should be acknowledged. Firstly, the evaluation of the effects of PQQ treatment on the thyroid and intestine lacks comprehensiveness. Additionally, the sample size and animal model employed in our study are small and single. Finally, the underlying mechanisms of PQQ supplement between thyroid and gut microbiota are unclear. Therefore, further investigations incorporating larger sample sizes, clinical research, and comprehensive analyses are imperative for future studies.

In summary, the supplementation of PQQ regulates thyroid function and mitigates thyroid injury, as well as reduces inflammation and oxidative stress response, thereby attenuating small intestine epithelial injury. Moreover, in microbial analyses, PQQ restores the diversity of gut microbiota and maintains the bacteria composition, initially impaired by GD. Notably, PQQ changes the primary composition and abundance of GD's intestine microbiota by moderating *Lactobacillus* and *Lactobacillus* may exert in the pathogenesis and progression of GD.

Availability of data and material

The original data presented in the study are available from the corresponding author on reasonable request.

Author contributions

All authors participated in the design of the study, and met criteria for authorship. Z.L., W.Y. and L.C. contributed to the research concept, experimental design, and project supervision. T.Q. and D.G. developed the mouse model and performed the H&E staining. M.Z. and H.C. conducted Real-Time PCR. X.L., W.J. and G.L. analyzed the data and wrote the paper. X.L., W.J. and G.L. contributed equally to this work and shared the co-first authorship. All authors critically reviewed and commented on the final manuscript.

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Conflict of interest

The authors do not report any financial or personal connections with other persons or organizations, which might negatively affect the contents of this publication and/or claim authorship rights to this publication.

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